Questions for the tutor class:

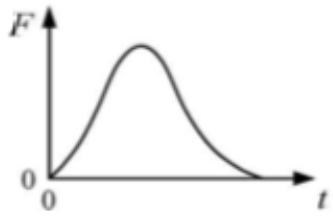
1, 2, 3, 4, 8, 9, 11, 12, 13, 15, 16, 18, 20, 22, 25

1. Which of the following is NOT one of the Newton's laws of motion?
   - **A. The total momentum of a system of interacting bodies remains constant, providing no external force acts.**
   - B. The rate of change of momentum of a body is directly proportional to the external force acting on the body and takes place in the direction of the force.
   - C. If body A exerts a force on body B, then body B exerts an equal and oppositely-directed force on body A.
   - D. A body continues in a state of rest or of uniform motion in a straight line unless acted upon by some external force.

2. A tractor of mass 1000 kg tows a trailer of mass 1000 kg. The total resistance to motion has a constant value of 4000 N and one quarter of this resistance acts on the trailer. The tractor and trailer are moving with with a constant speed of 6 m s^{-1}.
   What is the force exerted on the tractor by the tow-bar?
   - A. 0 N
   - **B. 1000 N**
   - C. 3000 N
   - D. 4000 N

3. A body is falling freely under gravity. What is its rate of change of momentum equal to?
   - A. impulse
   - B. kinetic energy
   - C. power
   - **D. weight**

4. The given graph shows the variation with time t of the force F exerted on a body. The area under the graph represents the change of a physical quantity associated with the body. What is this quantity?



   - A. acceleration
   - B. kinetic energy

C. momentum

D. potential energy

5. A box of mass 2 kg travels at 3 m/s constant speed when it is being pushed horizontally on a rough surface by a 4 N force. What is the acceleration of the box when the force increased to 6 N ?

   A. 0 m/s^2

   B. 1 m/s^2

   C. 2 m/s^2

   D. 4 m/s^2

6. A box is being pushed across a horizontal ground at constant speed. If the push is stopped, the box will

   A. stop immediately

   B. continue to move at constant speed

   C. start slowing down

   D. move at constant speed for a while and then slow down

7. A box is being pushed horizontally along a rough surface and constant acceleration is archived. If the push is stopped, the box will

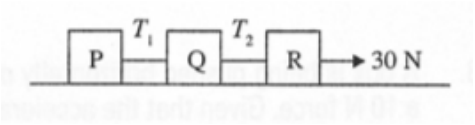
   A. slow down and finally stop

   B. stop immediately

   C. continue to move at constant speed

   D. accelerate at a slower rate

8. P, Q and R are three identical blocks resting on a surface. A force of 30 N is applied at one end as shown in the diagram below. What are the tension T_1, between blocks P and Q, and T_2, between blocks Q and R, in the string



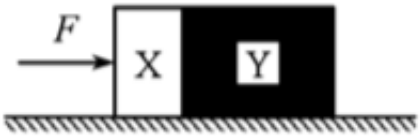
   A. $T_1 = 15$ N; $T_2 = 25$ N

   B. $T_1 = 30$ N; $T_2 = 30$ N

   C. $T_1 = 20$ N; $T_2 = 10$ N

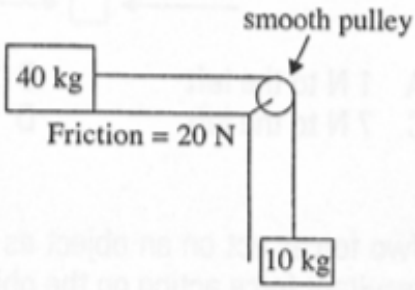
   D. $T_1 = 10$ **N;** $T_2 = 20$ **N**

9. Two blocks, X and Y, have masses m and $3m$ respectively. The given diagram shows the blocks being accelerated along a smooth horizontal surface by a force F applied to block X.
   During the acceleration, what is the magnitude of the force exerted by block X on block Y?



A. $F/4$
B. $F/3$
C. $F/2$
D. $3F/4$

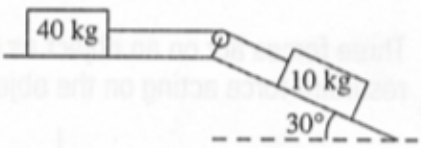
10. A 40 kg mass is initially at rest on a rough bench and a 10 kg mass is attached to it as shown. A friction force of 20 N opposes the motion of the 40 kg mass



What is the speed of the 10 kg mass 2 s after the masses start moving? (g=10 m/s^2)

A. 0 m/s
B. 3.2 m/s
C. 4.0 m/s
D. 20 m/s

11. A 40 kg mass and a 10 kg mass rest on smooth surface as shown



What is the speed of the 10 kg mass 2 s after the system is released from rest? (g = 10 m/^{2})

A. 1 m/s
B. 2 m/s
C. 4 m/s

D. 10 m/s

12. A lift moves downwards at a constant acceleration of 2 m/s^2. Given that the lift has a mass of 500 kg, what is the tension in the cable? (g = 10 N/kg)

    A. 2500 N

    B. 4000 N

    C. 5000 N

    D. 6000 N

13. A man of mass 50 kg stands on a weighting scale in a lift. When the lift is descending at a constant deceleration of 1.0 m/s^2, what is the reading on the weighting scale? (g= 10 N/kg)

    A. 60 N

    B. 450 N

    C. 500 N

    D. 550 N

14. A man of mass 50 kg stands on a weighting scale in a lift. When the lift is descending at a constant acceleration of 1.0 m/s^2, what is the reading on the weighting scale? (g= 10 N/kg)

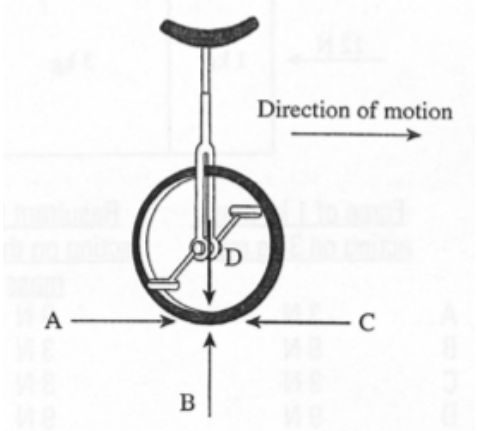
    A. 60 N

    B. 450 N

    C. 500 N

    D. 550 N

15. A unicycle is being peddled and is moving in the direction shown. which force shows the frictional force acting on the wheel?

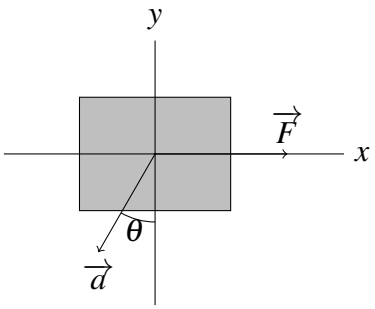


Solution: A

16. A motorist whose car will not start seeks a tow from a second motorist. The towing vehicle accelerates slowly from rest with the tow-rope slack; when the rope become tight, the towed car starts to accelerate and the towing vehicle moves with a constant speed of 0.8 m/s until the towed car, of mass 1000 kg, achieves the same speed.
    (a) What is the change of the momentum of the towed car?
    (b) The average tension in the rope may be assumed to be 4000 N during the acceleration of the towed car. How far does the towed car travel from rest before it reaches the speed of 0.8 m/s?
    (c) How far does the towing vehicle travel in this time?
    (d) The breaking strain of the towed-rope is 0.04. What must be the minimum unstretched length of the rope if the tow is accomplished without breakage of the rope?

Solution:
(a) 800 $kg.m/s$
(b) 0.08 m
(c) 0.16 m
(d) 2.0 m

17. There are two forces on a 2.00 kg box in the overhead view of the Figure below, but only one is shown. For $F_1 = 20.0\ N$, $a = 12.0\ m/s^2$, and $\theta = 30.0^\circ$. Find the second force:
    (a) in unit-vector notation,
    (b) a magnitude,
    (c) an angle relative to the positive direction of the x axis.



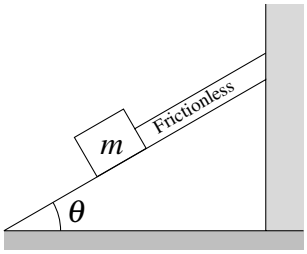
Solution:

(a) $\vec{F}_2 = -32\vec{i} - 21\vec{j}$

(b) $F_2 = 38$ N

(c) 213°

18. In the Figure below, let the mass of the block be $8.5kg$ and the angle θ be 30°. Find:
    (a) the tension in the cord,
    (b) the normal force acting on the block,
    (c) If the cord is cut, find the magnitude of the resulting acceleration of the block.



Solution:

(a) 42 N

(b) 72 N

(c) 4.9 m/s^2

19. A block is projected up a frictionless inclined plane with initial speed $v_0 = 3.50\ m/s$. The angle of incline is $\theta = 32.0^\circ$.
    (a) How far up the plane does the block go?
    (b) How long does it take to get there?
    (c) What is its speed when it gets back to the bottom?

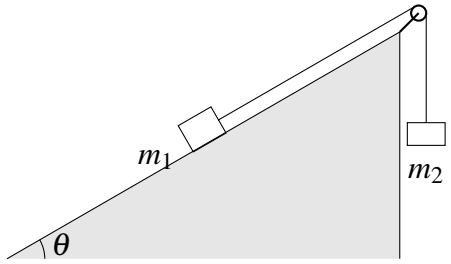
Solution:

(a) 1.18 m

(b) 0.674 s

(c) 3.50 m/s ($v = -3.50$ m/s)

20. A block of mass $m_1 = 3.70\ kg$ on a frictionless plane inclined at angle $\theta = 30.0^\circ$ is connected by a cord over a massless frictionless pulley to a second block of mass $m_2 = 2.30\ kg$. What are
    (a) the magnitude of the acceleration of each block,
    (b) the direction of the acceleration of the hanging block,
    (c) the tension in the cord?



Solution:

(a) $a_1 = a_2 = g(m_2 - m_1 \sin\theta)/(m_1 + m_2)$

(b)

(c) $T = m_1 m_2(1 + \sin\theta)/(m_1 + m_2)$

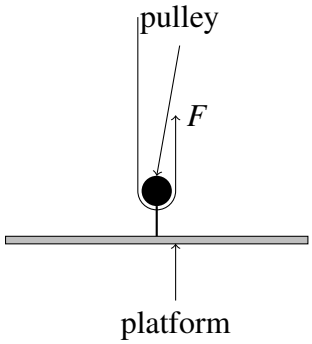
21. A certain force gives an object of mass m_1 an acceleration of 12.0 m/s^2 and an object of mass m_2 an acceleration of 3.30 m/s^2. What acceleration would the force give to an object of mass
    (a) $m_2 - m_1$,
    (b) $m_2 + m_1$?

Solution:

(a) 4.55 m/s^2

(b) 2.59 m/s^2

22. Consider the diagram. The platform and the attached frictionless pulley weigh a total of $34N$. With what force F must the (light) rope be pulled in order to lift the platform at 3.2 m/s^2?



Solution: $F = 22.55N$

23. A hockey puck is hit on a frozen lake and starts moving with a speed of 12 (m/s). Exactly 5.0 (s) later, its speed is 6.0 (m/s).
    (a) What is its average acceleration?
    (b) What is the coefficient of kinetic friction between the puck and the ice?
    (c) How far does the puck travel during this 5.0 (s) interval?

Solution:
(a) -1.2 m/s^2
(b) 0.12
(c) 45 m

24. A 3.00 (kg) block starts from rest at the top of a 30.0° incline and accelerates uniformly down the incline, moving 2.00 (m) in 1.50 (s).
    (a) Find the magnitude of the acceleration of the block.
    (b) Find the coefficient of kinetic friction between the block and the incline.
    (c) Find the magnitude of the frictional force acting on the block.
    (d) Find the speed of the block after it has slid a distance of 2.00 (m)

Solution:

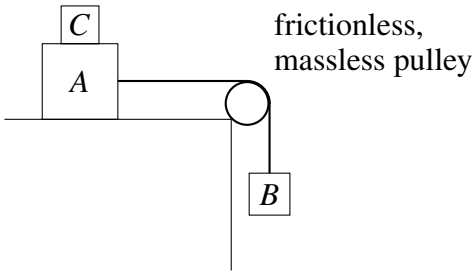
(a) 1.78 m/s^2

(b) 0.37

(c) 9.4 N

(d) 2.67 m/s

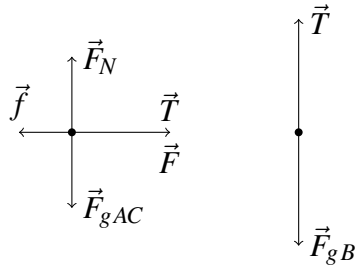
25. Blocks A and B have weights of 44 N and 22 N, respectively.



(a) Determine the minimum weight of block C to keep A from sliding if μ_s between A and the table is 0.2.

(b) Block C suddenly is lifted off A. What is the acceleration of block A if μ_K between A and the table is 0.15.

Solution:



(a) Assume the blocks are not moving. For the blocks on the table we take the x axis to be to the right and the y axis to be upward. From Newton's second law, we have: $T = f$ and $F_N = W_{AC}$

For block B take the downward direction to be positive. Newton's second law for that block B is: $W_B = T$

Combine the above equations, we have: $f = T = W_B$ and $F_N = W_{AC}$

If sliding is not to occur, f must be less than $\mu_S F_N$ or $W_B < \mu_S W_{AC}$. The smallest that W_{AC} can be with the blocks still at rest is: $W_{AC} = \dfrac{W_B}{\mu_S} = 110N$

Since the weight of block A is $44N$, the least weight for C is $(110-44)N = 66N$.

(b) The Newton's second law equations become:

$$\begin{aligned} T - f &= (W_A/g)a, \\ F_N &= W_A, \\ W_B - T &= (W_B/g)a \end{aligned}$$

where $f = \mu_K.F_N$
Combine the above equations, we have :

$$a = \frac{g(W_B - \mu_K W_A)}{W_A + W_B} = 2.3m/s^2$$